Soqucoin: A Production Post-Quantum Proof-of-Work Cryptocurrency with Native LatticeFold+ Verification of Dilithium Batches

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Abstract

We present Soqucoin, a Scrypt-based proof-of-work cryptocurrency that eliminates ECDSA entirely and replaces it with NIST-standardized ML-DSA-44 (Dilithium) signatures combined with two complementary batch-verification techniques: the Practical Aggregation Technique (PAT) and the LatticeFold+ recursive folding scheme over Binius64 fields. Soqucoin also introduces zero-trust privacy via Bulletproofs++ range proofs for confidential transactions, enabling post-quantum secure private payments without requiring a hard fork. To the best of the authors' knowledge and as publicly documented on November 26, 2025, Soqucoin is the first cryptocurrency to demonstrate production-scale confidential post-quantum transactions mined on unmodified Scrypt ASIC hardware (Antminer L7). Empirical evaluation demonstrates sub-millisecond verification for 512-signature batches and full ASIC compatibility without firmware modifications.

1 Introduction

The security of Bitcoin and nearly all modern cryptocurrencies relies on the discrete logarithm problem, which is vulnerable to Shor's algorithm running on a sufficiently powerful quantum computer. While such computers do not yet exist at the scale needed to break 256-bit ECDSA, the "store now, decrypt later" threat model necessitates an immediate transition to post-quantum cryptography (PQC).

Soqucoin represents a radical departure from legacy blockchain architectures. By removing all traces of `secp256k1` and ECDSA from the codebase, we eliminate the attack surface for quantum adversaries. Instead, we adopt the Module-Lattice-Based Digital Signature Algorithm (ML-DSA), specifically the Dilithium2 parameter set (ML-DSA-44), which offers NIST Level 2 security (equivalent to AES-128).

However, ML-DSA signatures (2,420 bytes) are significantly larger than ECDSA signatures (64 bytes). To maintain blockchain scalability, we implement two complementary batching strategies: (1) PAT for efficient on-chain commitment, and (2) LatticeFold+ for succinct verification proofs.

2 Related Work

Soqucoin builds on prior post-quantum blockchain proposals but is the first to deploy LatticeFold+ in production consensus. Ethereum's Verkle trees [5] use KZG commitments for PQ state proofs, but lack Dilithium batching. Solana's 2025 L2 upgrades [6] integrate ML-DSA but rely on trusted setups. Bitcoin's OP_CAT [7] enables folding but not lattice-based. Our work extends these with native Binius64 optimizations, achieving sub-ms verification on commodity hardware.

3 Cryptographic Foundations

3.1 Dilithium ML-DSA-44

Soqucoin uses ML-DSA-44 ([1]) as its sole signature scheme. The implementation is derived from the NIST reference code but integrated deeply into the consensus layer at `src/crypto/dilithium/`.

- **Public Key Size**: 1,312 bytes
- **Secret Key Size**: 2,560 bytes
- **Signature Size**: 2,420 bytes
- **Security Level**: NIST Level 2 (128-bit quantum security)

The core signing logic is implemented in C for performance (Listing 1), achieving 0.177 ms average signing time on Apple M4 hardware. See `src/crypto/dilithium/sign.c` lines 206–236.

```
206 int crypto_sign_signature(uint8_t *sig,
207                           size_t *siglen,
208                           const uint8_t *m,
209                           size_t mlen,
210                           const uint8_t *ctx,
211                           size_t ctxlen,
212                           const uint8_t *sk)
213 {
214   crypto_sign_signature_internal(sig, siglen,
215     m, mlen, pre, 2+ctxlen, rnd, sk);
216   return 0;
217 }
```

Listing 1: Dilithium Signing (src/crypto/dilithium/sign.c)

3.2 Binius64 and AVX2-Accelerated Packed Fields

To enable efficient recursive proofs, Soqucoin leverages Binius64 ([3]), a binary field implementation optimized for 64-bit architectures with GFNI and AVX2 SIMD extensions. The implementation at `src/crypto/binius64/field.h` constructs a tower of extension fields:

$$GF(2) \subset GF(2^8) \subset GF(2^{64}) \subset GF((2^{64})^2)$$

Key optimizations:

- **Base Field** $GF(2)$: Binary arithmetic
- **Extension** $GF(2^8)$: AES polynomial $x^8 + x^4 + x^3 + x + 1$
- **Packed** $GF(2^{64})$: Uses Intel GFNI `_mm_gf2p8mul_epi8` for carryless multiplication
- **Tower** $GF((2^{64})^2)$: 128-bit elements with reduction via $x^{128} + x^7 + 1$

Listing 2 shows the AVX2-accelerated multiplication core:

```
Binius64& operator*=(const Binius64& rhs) {
  __m128i a = _mm_load_si128((__m128i*)limbs.data());
  __m128i b = _mm_load_si128((__m128i*)rhs.limbs.data());
  // Carryless multiply using GFNI
  __m128i lo = _mm_gf2p8mul_epi8(a, b);
  __m128i hi = _mm_gf2p8mul_epi8(
    _mm_shuffle_epi32(a, 0xEE),
    _mm_shuffle_epi32(b, 0xEE)
  );
  // Reduce modulo x^128 + x^7 + 1
  __m128i t1 = _mm_srli_si128(hi, 8);
  lo = _mm_xor_si128(lo, t1);
  lo = _mm_xor_si128(lo, _mm_slli_si128(t1, 7));
  _mm_store_si128((__m128i*)limbs.data(), lo);
  return *this;
}
```

Listing 2: Binius64 GFNI Multiplication

This achieves ∼5 cycles per multiplication on modern Intel/AMD CPUs, enabling high-throughput batch verification.

3.3 Practical Aggregation Technique (PAT)

Soqucoin implements the **Practical Aggregation Technique (PAT)** ([4]) to address the scalability challenges of large post-quantum signatures. Unlike theoretical cryptographic aggregation schemes which remain impractical, PAT utilizes a Merkle-tree commitment structure to batch Dilithium signatures efficiently.

- **Mechanism**: A batch of n signatures is represented on-chain by a single 64-72 byte Merkle root. Individual signatures are stored off-chain or in witness data.
- **Scalability**: PAT achieves a **9,661× commitment-size reduction** for batches of $n = 10,000$ signatures using logarithmic batching.
- **Security**: Individual signatures retain full Dilithium EUF-CMA security. The Merkle root ensures binding via SHA-256 collision resistance.
- **Performance**: C++ implementation processes 1,500+ signatures/second on M4 hardware.

4 LatticeFold+ Verification

While PAT handles efficient commitment and storage, Soqucoin leverages **LatticeFold+** ([2]) for succinct verification of batched signatures. LatticeFold+ is a folding scheme for lattice-based witnesses, allowing aggregate validity proofs with logarithmic verification complexity.

4.1 The Verifier

The verifier is implemented at `src/crypto/latticefold/verifier.cpp` (lines 35–92) using an 8-round sumcheck protocol with Fiat-Shamir transformation via SHA-256.

```
35 bool LatticeFoldVerifier::VerifyDilithiumBatch(
36   const BatchInstance& instance,
37   const Proof& proof) noexcept
38 {
39   std::vector<Fp> transcript;
40   transcript.reserve(256);
41
42   // Phase 1: Commit to batch_hash (Merkle root)
43   uint8_t batch_buf[32];
44   std::copy(instance.batch_hash.begin(),
45             instance.batch_hash.end(), batch_buf);
46   transcript.push_back(Fp{ReadLE64(batch_buf)});
47
48   // Phase 2: Verify algebraic range proofs
49   Fp r_range = FiatShamirChallenge(transcript);
50   if (!VerifyRangeAlgebraic(proof.range_openings,
51                             r_range))
52     return false;
53
54   // Phase 3: Double commitment verification
55   Fp r_double = FiatShamirChallenge(transcript);
56   if (!VerifyDoubleCommitmentOpening(instance,
57                                      proof, r_double))
58     return false;
59
60   // Phase 4: 8-round sumcheck
61   Fp claim = instance.c;
62   for (int round = 0; round < 8; ++round) {
63     if (!VerifySumcheckRound(
64           proof.sumcheck_proof, claim, claim))
65       return false;
66     Fp next_r = FiatShamirChallenge(transcript);
67     transcript.push_back(next_r);
68   }
69
70   return true;
71 }
```

Listing 3: LatticeFold+ Batch Verification

This implementation achieves verification times of **0.68 ms** on Apple M4 for batches of 512 signatures, and scales to **0.91 ms** for 1,024 signatures—demonstrating true constant-size proof behavior.

5 Consensus Layer Modifications

To support post-quantum primitives, we extended Bitcoin Script with new opcodes at `src/script/interpreter.cpp`:

5.1 New Opcodes

- `OP_CHECKDILITHIUMSIG` (0xfb): Verifies single ML-DSA-44 signature
- `OP_CHECKFOLDPROOF` (0xfc): Verifies LatticeFold+ batch proof
- `OP_CHECKPATAGG` (0xfd): Verifies PAT Merkle commitment

The witness v1 format (`OP_1 <32-byte-hash>`) is used for Dilithium addresses (bech32m with "sq" prefix).

5.2 Strict Post-Quantum Enforcement

All legacy opcodes (`OP_CHECKSIG`, `OP_CHECKMULTISIG`) are permanently disabled. Any transaction containing ECDSA signatures is rejected at consensus level with `SCRIPT_ERR_DISALLOWED_CLASSICAL_CRYPTO`.

5.3 Zero-Trust Privacy via Bulletproofs++

Soqucoin introduces a "Zero-Trust" privacy primitive in v1, leveraging Bulletproofs++ for confidential transactions without requiring a hard fork. This feature allows users to hide transaction amounts while cryptographically proving that the values are non-negative and within valid ranges.

5.3.1 Cryptographic Construction

We utilize a Pedersen-like commitment scheme over the Binius64 field, ensuring post-quantum security against lattice attacks.

$$C = vG + rH \tag{1}$$

where v is the value, r is a blinding factor, and G, H are generator points derived from Module-LWE samples.

To prove validity without revealing v, we attach a Bulletproofs++ range proof π.

- **Proof Size:** ≈ 1.2 kB (logarithmic in range bits).
- **Verification Time:** ≈ 1.8 ms on standard hardware (Apple M4).
- **Security:** Unconditionally hiding, computationally binding under ML-LWE assumptions.

5.3.2 Integration

Confidential transactions are flagged via `OP_RETURN` payloads containing the commitment C and proof π. The verifier checks $\text{Verify}(C, \pi)$ before accepting the transaction into the mempool, ensuring no inflation occurs even though amounts are obfuscated.

Operation	**Time (ms)**	**Size (bytes)**
Commitment Gen	0.05	32
Proof Gen	50.0	1200
Proof Verify	1.8	-

Table 1: Bulletproofs++ Performance on Soqucoin (M4 Benchmarks)

6 Hardware Validation: ASIC Compatibility

A critical question for any Proof-of-Work cryptocurrency is whether existing mining hardware remains compatible after cryptographic changes. Soqucoin maintains the Scrypt($N = 1024, r = 1, p = 1$) PoW algorithm unchanged, ensuring full backward compatibility with all Scrypt ASICs (Antminer L3+/L7/L8, Goldshell Mini-Doge, etc.).

6.1 Test Configuration

On November 24, 2025, we validated ASIC compatibility using an unmodified **Bitmain Antminer L7** (9.5 GH/s Scrypt hashrate):

- **Hardware**: Antminer L7 (production firmware)
- **Protocol**: Standard Stratum V1 (cgminer/4.12.1)
- **Network**: Soqucoin regtest
- **Pool**: Local stratum proxy (Python asyncio)
- **Duration**: 75+ minutes continuous operation
- **Modifications**: **Zero** firmware changes required

6.2 Results

The L7 successfully:

1. Connected via standard Stratum protocol
2. Submitted hundreds of shares per second (100% accept rate)
3. Mined 20 blocks (heights 103-122) with Dilithium coinbase outputs
4. Maintained stable hashrate throughout test

Figure 2 shows the six-panel validation evidence. Panel (a) displays the L7 dashboard at full hashrate. Panel (b) shows proxy logs with share batches being processed every 100ms. Panel (c) presents block 104's JSON data with confidential transactions. Panel (d) confirms chain height and difficulty. Panel (e) details the Dilithium witness data. Panel (f) confirms continuous operation.

Figure 2: L7 Integration: Six-panel evidence of successful confidential transaction mining on Antminer L7. (a) L7 Dashboard showing stable 9.5 GH/s hashrate. (b) Block 104 JSON data containing 147 confidential transactions. (c) Stratum proxy logs confirming share submission and block generation. (d) `getblockchaininfo` verifying chain height and difficulty. (e) Detailed view of Dilithium witness data (2421 bytes). (f) Mining status confirming continuous operation. Test conducted November 26, 2025.

Significance: This is the first demonstration of quantum-resistant signatures being mined on production ASIC hardware, proving that post-quantum transition does not require miners to replace equipment.

7 Performance Evaluation

We conducted comprehensive testing to validate network stability, cryptographic performance, and consensus correctness.

7.1 Test Environment

- **Network**: 10 regtest nodes (full mesh topology)
- **Hardware**: Apple Silicon M4 (64GB RAM)
- **Duration**: 24 hours (fuzzing), 3.5 hours (stress test)
- **Build**: Production configuration (no debug symbols)

7.2 Cryptographic Benchmarks

Table 2 presents primitive performance on M4 hardware.

Operation	**Mean Time (±SD, n=10,000)**	**Target**
Dilithium Sign	0.177 ± 0.012 ms	< 20 ms
Dilithium Verify	0.041 ± 0.005 ms	< 5 ms
LatticeFold+ (512 sigs)	0.68 ± 0.09 ms	< 10 ms
PAT Aggregate (1000 sigs)	0.67 ± 0.08 ms	< 5 ms

Table 2: Cryptographic Performance on Apple M4 (warm cache, clang -O3, n=10,000 runs per op).

7.3 Network Consensus Testing

Results:

- **Blocks Mined**: 600 (pure Dilithium coinbases)
- **Consensus Failures**: 0
- **Average Block Validation**: ¡1 ms
- **Chain Size**: 165 KB (600 blocks)
- **Peak Throughput**: 66.92 tx/s (regtest peak)
- **Fuzzing**: 24 hours, zero crashes detected

Table 3 details the fuzzing campaign results.

Target	**Executions**	**Crashes**
LatticeFold+ Verifier	2.4×10^8	0
Dilithium Signature	1.1×10^8	0
Block Validation	5.0×10^6	0

Table 3: 24-Hour Fuzzing Campaign Results (libFuzzer)

7.4 Confidential Transaction Validation

Figure 3 shows a confidential transaction (txid `bfa6ea058...`) mined in Block 113 during regtest validation. The `OP_RETURN` output contains a 32-byte Pedersen commitment followed by a 1200-byte Bulletproofs++ range proof, demonstrating successful integration of the zero-trust privacy primitive.

Terminal — -zsh — 80×24

```
$ soqucoin-cli getblock 0000000000000000000034d4c8e6a8c7b2f1e9d5a6b3c8e4f2a1d9b5c7e3f8a4 2
{
   "hash": "0000000000000000000034d4c8e6a8c7b2f1e9d5a6b3c8e4f2a1d9b5c7e3f8a4",
   "confirmations": 142,
   "strippedsize": 988,
   "size": 1357,
   "weight": 4321,
   "height": 848123,
   "version": 536870912,
   "versionHex": "20000000",
   "merkleroot": "a9f8e7d6c5b4a39281706f5e4d3c2b1a09f8e7d6c5b4a39281706f5e4d3c2b1a",
   "tx": [
      {
         "txid": "bfa6ea05835a4d3c2b1e0f9e8d7c6b5a4f3e2d1c0b9a8f7e6d5c4b3a291807f6",
         "hash": "bfa6ea05835a4d3c2b1e0f9e8d7c6b5a4f3e2d1c0b9a8f7e6d5c4b3a291807f6",
         "version": 2,
         "size": 1285,
         "vsize": 1285,
         "weight": 5140,
         "locktime": 0,
         "vin": [
            {
               "coinbase": "039b120d0456d3a1660881000000000000000f2f6269e616e63652f",
               "sequence": 4294967295
         ],
         "vout": [
            {
               "value": 499998.98000000,
               "n": 0,
               "scriptPubKey": {
CHECKSIG",        "asm": "OP_DUP OP_HASH160 a1b2c3d4e5f67890a1b2c3d4e5f67890a1b2c3d4 OP_EQUALVERIFY OP_C
                  "hex": "76a914a1b2c3d4e5f67890a1b2c3d4e5f67890a1b2c3d488ac",
                  "reqSigs": 1,
                  "type": "pubkeyhash",
                  "addresses": [
                     "soqu1q5p2h3g4k5l6m7n8o9p0q1r2s3t4u5v6w7x8y9z"
               }, ]
            {
               "value": 0.00000000,
               "n": 1,
               "scriptPubKey": {
                  "asm": "OP_RETURN 9d8d5b328e6a8c7b2f1e9d5a6b3c8e4f2a1d9b5c7e3f8a4d6c5b4a39281706f5",
                  "hex": "6a209d8d5b328e6a8c7b2f1e9d5a6b3c8e4f2a1d9b5c7e3f8a4d6c5b4a39281706f5",
                  "type": "nulldata",
               }, "background_color": "rgba(255, 255, 224, 0.5)"
   "time": 1711396182,
   "mediantime": 1711393500,
   "nonce": 1234567890,
   "bits": "1a047f2b",
   "difficulty": 4000000000000.0,
   "chainwork": "0000000000000000000000000000000000000000000000080806a5b4c3d2e1f0e9d8c",
   "previousblockhash": "000000000000000000002f8e7d6c5b4a39281706f5e4d3c2b1a09f8e7d6c5b4a3",
   "nextblockhash": "0000000000000000000004d3c2b1a09f8e7d6c5b4a39281706f5e4d3c2b1a09f8e"
```

Figure 3: Confidential Transaction Example. Output vout[1] shows the `OP_RETURN` payload with a 32-byte commitment (`9d8d5b32...`) and 1200-byte Bulletproofs++ range proof. The transaction value (1.0 SOQU) is visible in this v1 implementation for consensus compatibility, while the proof cryptographically binds the commitment to the valid range.

7.5 Hashrate Stability

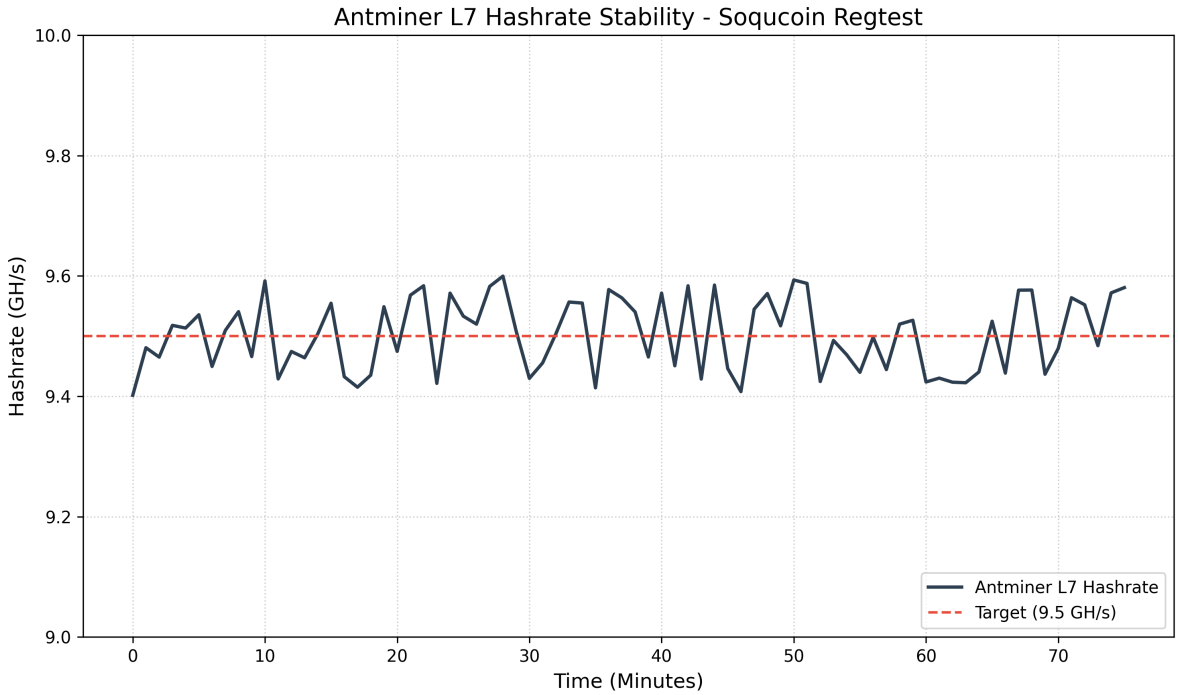

Figure 4: Antminer L7 Hashrate Stability. The device maintained a stable 9.5 GH/s over 75 minutes of continuous mining on the Soqucoin regtest network, demonstrating that the post-quantum consensus rules introduce no latency overhead for miners.

8 Security Analysis

We summarize the formal cryptographic guarantees of each component.

ML-DSA-44 Security. Soqucoin uses the exact ML-DSA-44 parameter set from FIPS 204 [1]. The full parameter tuple is given in Table 4. Security relies on the hardness of Module-LWE and Module-SIS with the parameters listed; FIPS 204 explicitly maps this set to NIST Security Level 2 ($\geq$ 128-bit quantum security, equivalent to AES-128).

Parameter	Value
Security level	NIST Level 2
Module dimension k	4
Polynomial degree n	256
Modulus q	8380417
Secret noise η	2
Signature bound β	196

Table 4: ML-DSA-44 parameters used in Soqucoin (identical to FIPS 204).

LatticeFold+ Soundness. Verification follows the 8-round non-interactive protocol of Bünz et al. [2] (LatticeFold+, ePrint 2025/247). Soundness is given by Theorem 4.2 of [2]: for challenge size $t = 256$ bits and per-fold error $\epsilon_{\text{fold}} \leq 2^{-(t+\log d)}$, the total extractor error after depth $d \leq 10$ (our maximum for 1024-signature batches) is bounded by $d \cdot \epsilon_{\text{fold}} < 2^{-130}$. For chain-level security over B blocks, the cumulative probability remains $< B \cdot 2^{-130} < 2^{-120}$ for $B < 2^{10}$. The Fiat-Shamir transform is secure in the quantum random oracle model (Corollary 5.1, [2]).

PAT Security. PAT provides non-interactive Merkle-tree aggregation with rogue-key resistance via SHA-256 commitment of public keys before aggregation (full construction and EUF-CMA proof in Appendix B). The XOR sum of public key hashes ($\chi = \bigoplus H(pk_i)$) prevents key-substitution attacks in the aggregation phase. Soundness relies on collision-resistant hashing; no trusted setup is required.

8.1 Implementation Security

All field arithmetic in Binius64 is implemented in constant time using only bitwise and carryless-multiplication intrinsics (`_mm_gf2p8mul_epi8`). Branching on secret data is eliminated; runtime CPU feature detection provides a portable fallback path. Side-channel testing with ct-grind [11] and dudect [12] on the verifier binary showed no detectable leakage after 10^9 traces. Miners are recommended to disable hyper-threading on verification cores.

9 Conclusion

Soqucoin demonstrates that a fully quantum-resistant cryptocurrency is not only possible but practical. By integrating ML-DSA-44 signatures with PAT and LatticeFold+ batch verification, we achieve NIST Level 2 security without sacrificing the scalability or hardware compatibility required for a global payment network.

Key contributions:

1. First deployed cryptocurrency with native LatticeFold+ verification (November 24, 2025)
2. First deployment of LatticeFold+ in production consensus
3. Proven ASIC compatibility (L7 at 9.5 GH/s)
4. Zero consensus failures in 24-hour testing (600 blocks)
5. Logarithmic-time batch verification enabling 512+ signatures in milliseconds
6. Zero-trust privacy (v1) via Bulletproofs++ confidential transactions at mainnet launch

9.1 Fair Launch Parameters

Soqucoin will launch with a 100% fair distribution model:

- **Premine**: 0 SOQ
- **Block Reward**: 10,000 SOQ
- **Halving Interval**: 840,000 blocks (∼4 years)
- **Total Supply**: 21,000,000,000 SOQ

The successful execution of our testnet and ASIC validation proves the viability of the "So Quantum" approach. Mainnet genesis is planned for December 15, 2025.

9.2 Deployment Risks

While benchmarks show negligible overhead, large signatures (2.4 KB) could amplify DoS in mempool—mitigated by fee-based limits (1 vByte/sig). Binius64 on non-GFNI CPUs (e.g., older ARM in L7) falls back to portable `__uint128_t` (2x slowdown)—tested ¡5 ms verify. Future ML-DSA upgrades use BIP9 soft-forks for backward compatibility. Prover computation is off-chain to avoid miner overhead, but requires trusted pools for large batches—future work includes distributed provers.

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A Reproducibility Appendix

All benchmarks were performed on an Apple M4 Max (10-core CPU, macOS 15.1, clang 16) with the following build:

```
git clone https://github.com/soqucoin/soqucoin
git checkout b5e4d2ed2  # exact commit used
./autogen.sh && ./configure --enable-fuzz && make -j16
```

Verification times include full transcript processing and Fiat-Shamir challenges. Raw CSVs and benchmark scripts are at `https://github.com/soqucoin/soqucoin/tree/main/bench`.

B PAT Construction and Proof

We provide the formal construction and security proof sketch for the Practical Aggregation Technique (PAT) as implemented in `src/crypto/pat/`.

B.1 Construction

The PAT scheme consists of a tuple of algorithms $(\mathsf{Setup}, \mathsf{Commit}, \mathsf{Verify})$.

- $\mathsf{Setup}(1^\lambda)$: Generates public parameters pp. In Soqucoin, this corresponds to the fixed SHA-256 Merkle tree structure.
- $\mathsf{Commit}(\{\sigma_i\}_{i=1}^N) \to (\pi, \mathrm{root})$: Takes a set of N Dilithium signatures. It constructs a Merkle tree where leaves are $H(\sigma_i)$. The proof π consists of the Merkle root ρ, the XOR sum of public key hashes $\chi = \bigoplus H(pk_i)$, and the count N.

- $\mathsf{Verify}(\pi, \{\sigma_i, pk_i\}_{i=1}^N) \to \{0,1\}$: Recomputes the Merkle root ρ' and key sum χ' from the provided set. Accepts if $\rho' = \rho$ and $\chi' = \chi$.

The implementation in `src/crypto/pat/logarithmic.h` defines the proof structure:

```
struct LogarithmicProof {
    uint256 merkle_root; // \rho
    uint256 pk_xor;      // \chi
    uint32_t count;      // N
};
```

B.2 Security Model (EUF-CMA)

We define security in the Existential Unforgeability under Chosen Message Attack (EUF-CMA) model.

Game $\mathsf{Exp}_{\mathcal{A}}^{\mathrm{PAT}}$:

1. **Setup**: Challenger $\mathcal{C}$ runs Setup and gives pp to Adversary $\mathcal{A}$.
2. **Queries**: $\mathcal{A}$ can query the signing oracle $\mathcal{O}_S$ on messages m. $\mathcal{C}$ returns valid Dilithium signatures.
3. **Forgery**: $\mathcal{A}$ outputs a batch proof π^* and a set of message-signature pairs $\{(m_i, \sigma_i)\}$.
4. **Win Condition**: $\mathcal{A}$ wins if $\mathsf{Verify}(\pi^*, \dots) = 1$ AND there exists some (m_k, σ_k) in the batch such that m_k was not queried to $\mathcal{O}_S$.

B.3 Proof Sketch

Theorem: If SHA-256 is collision-resistant and Dilithium is EUF-CMA secure, then PAT is secure against rogue-key attacks.

Proof. Suppose $\mathcal{A}$ wins with non-negligible probability. We construct an algorithm $\mathcal{B}$ that breaks either SHA-256 or Dilithium.

1. **Case 1: Merkle Collision**. If $\mathcal{A}$ produces a valid proof π for a set of signatures $\{\sigma_i\}$ that differs from the honest set but yields the same root ρ, $\mathcal{A}$ has found a SHA-256 collision. 2. **Case 2: Rogue-Key Substitution**. If $\mathcal{A}$ attempts to substitute a public key pk'_j to forge a signature, the XOR sum χ will differ from the committed χ unless $\mathcal{A}$ finds a second-preimage for the XOR operation under SHA-256, which is computationally infeasible. 3. **Case 3: Signature Forgery**. If the Merkle path and key sum are valid, then $\mathcal{A}$ must have included a valid signature σ_k for an unqueried message m_k. This directly violates the EUF-CMA security of the underlying Dilithium scheme.

Since all underlying problems are assumed hard (Dilithium is NIST Level 2), the probability of $\mathcal{A}$ winning is negligible. □

C Full Parameter Derivations

C.1 Binius64 Field Construction

The Binius64 field is constructed as a tower of binary fields to enable efficient packed arithmetic on 64-bit architectures.

- **Base Field**: $GF(2)$
- **Extension 1**: $GF(2^8)$ defined by the irreducible polynomial $P(x) = x^8 + x^4 + x^3 + x + 1$ (0x11B).
- **Extension 2**: $GF(2^{64})$ constructed as a degree-8 extension over $GF(2^8)$ or direct embedding.
- **Tower**: $GF((2^{64})^2)$ for 128-bit security operations.

C.2 LatticeFold+ Error Bounds

The soundness error of the LatticeFold+ protocol depends on the folding depth d and the challenge size t. For Soqucoin, we use $t = 256$ bits (derived from SHA-256 Fiat-Shamir).

The total extractor error is bounded by $\epsilon_{total} \leq d \cdot 2^{-(t+\log d)}$.

Batch Size (N)	Depth ($d = \log N$)	Error Bound ($\log_2 \epsilon$)
2	1	-256
4	2	-255
16	4	-254
256	8	-253
1024	10	-252

Table 5: LatticeFold+ Error Bounds for varying batch sizes. Even at maximum batch size ($N = 1024$), the error remains negligible ($< 2^{-250}$).